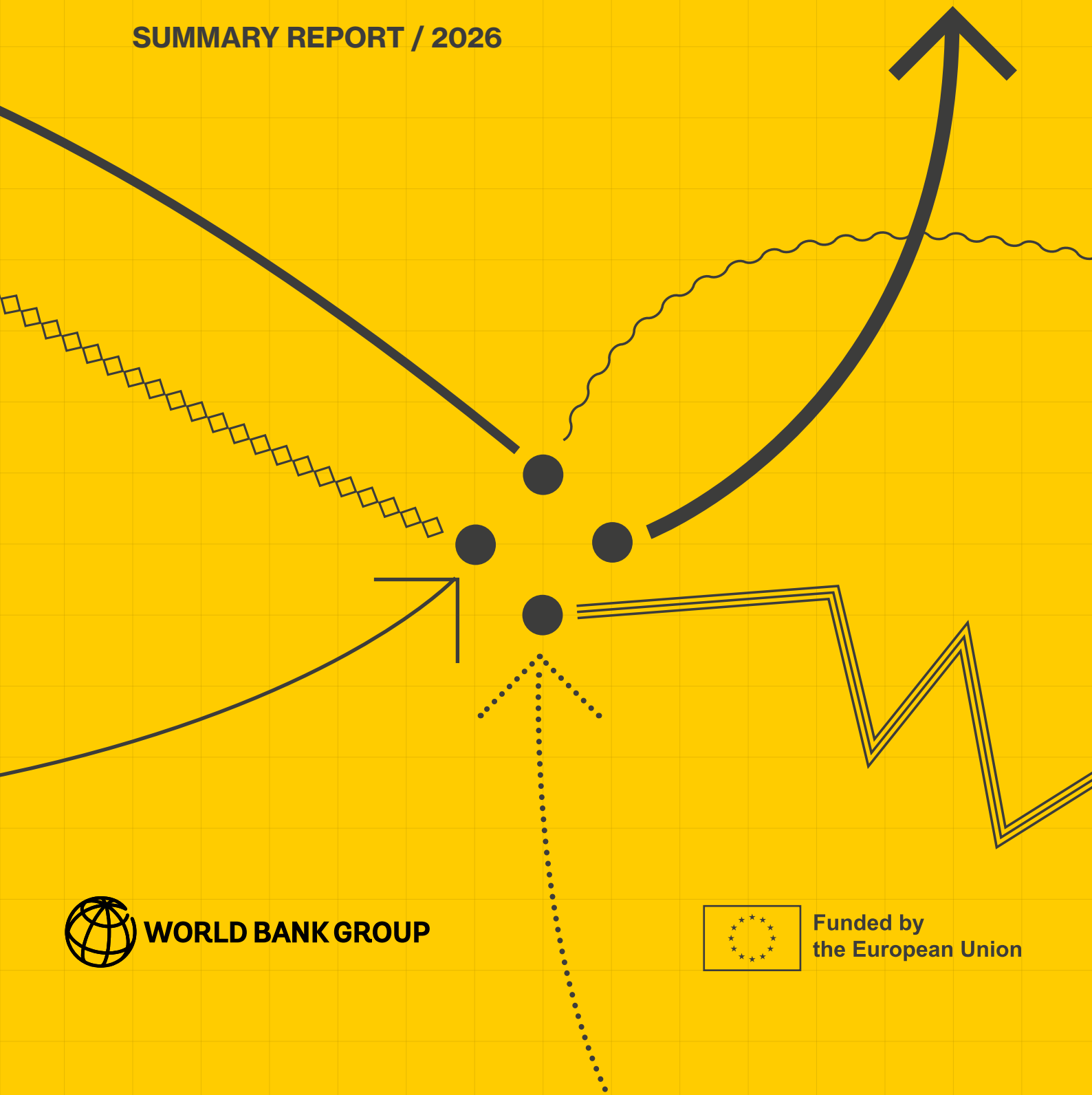
European Regional
Competitiveness Policy Lab

Short-Term Impacts of the First RCTs with Cohesion Policy

SUMMARY REPORT / 2026



WORLD BANK GROUP

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The Commission's Directorate-General
for Regional and Urban Policy

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London Business School

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Moravian-Silesian Innovation Centre

NUCLEUS

RIS3 Extremadura



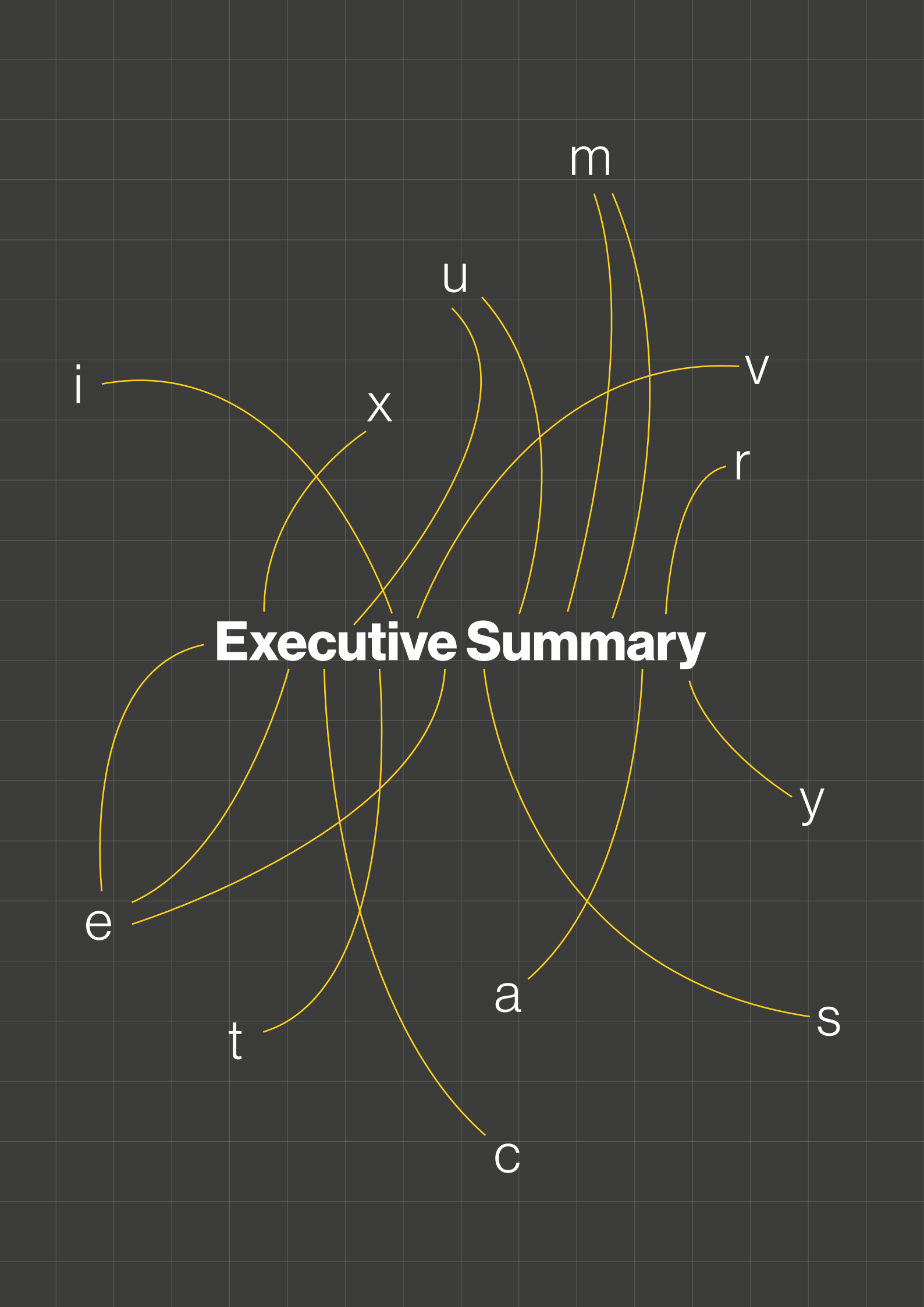


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AI	Artificial Intelligence
CRM	Customer Relationship Management
DESI	Digital Economy and Society Index
DG REGIO	Directorate-General for Regional and Urban Policy
EC	European Commission
EQA	European Quality Assurance
ERP	Enterprise Resource Planning
EU	European Union
FUNDECYT-PCTEx	Foundation for the Development of Science and Technology and the Extremadura Science and Technology Park Foundation
ICUK	Innovation Center Ústí nad Labem
ITT	Intention-to-treat
IE	Impact Evaluation
JTP	Just Transition Programme
JUNTA	La Junta de Extremadura, the Regional Government of Extremadura
KARP	Karlovarská Agentura Rozvoje Podnikání
Mbps	Megabits per second
MSIC	Moravian Silesian Innovation Centre
MSME	Micro, Small and Medium Sized Enterprise
PAP	Pre-Analysis Plan
RCT	Randomised Controlled Trial
RDA	Regional Development Agency
RDI	Research, Development, and Innovation
RIS3	Research and Innovation Smart Specialisation Strategy
SME	Small and Medium Enterprise


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Executive Summary
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Unlock Private-Sector Innovation and Raise Competitiveness

Innovation in policymaking to maximise returns on European Cohesion Policy investments is crucial for improving Europe's competitiveness and addressing global and local challenges. As global challenges grow, innovation must accelerate. Innovation is particularly vital for middle-income countries to avoid the middle-income trap and transition to high-income status by integrating advanced technologies, fostering new ideas, enhancing productivity, and ensuring global competitiveness. Moreover, innovation drives efficient use of capital, labor, and energy, which is essential for sustainable development, especially in light of rising debt, aging populations, and environmental challenges.

Significant funds are allocated to support the objectives of Cohesion Policy, making it one of the largest areas of EU investment. For 2021–2027, €392 billion was allocated to Cohesion Policy, out of a total EU budget of €1,2 trillion. For the next Multiannual Financial Framework (2028–2034), the European Commission has proposed an almost €2 trillion budget, with around €451 billion available for competitiveness, more than in the previous funding period. Approximately one-tenth of this amount is planned to be directed toward Research and Innovation. Additionally, the Recovery and Resilience Facility (RRF) was established in 2021 as a temporary instrument to aid recovery from the COVID-19 economic crisis, promoting sustainable growth and enhancing resilience against future shocks.

While the European Union is widely acknowledged as a major investor in innovation, significant regional disparities in innovation outcomes persist and limit its overall competitiveness. To remain on par with global counterparts, each Euro invested in innovation, research, and development must yield greater returns and have a more substantial impact on regional

economic growth. Smart Specialisation Strategies, which focus research and innovation policy interventions on areas of competitive advantage, are a central component of the Cohesion Policy framework. While innovation is a primary goal, the process to achieve it often lacks mechanisms for continuous learning and improvement. Successful policy development, particularly in innovation, must be accompanied by ongoing adaptation to evolving circumstances. Policymakers need to identify challenges, test innovative solutions, and refine approaches to address them. The evaluation community can assist by providing the necessary knowledge and expertise to support a trial-and-error policymaking approach, helping determine which policies are most effective.

A New Toolkit for Innovation in Policy-Making: Experimentation

Despite this large funding pool, knowledge about the causal impacts of Cohesion Policy programmes and their implementation approaches is limited. For the European Union (EU) to compete globally, it must ensure that the funds Member States invest yield substantial returns, which requires continuous policy improvement. For the 2021-2027 period, learning how to invest this funding (cost-)effectively is essential to strengthen policies that aim to create jobs, promote just transitions, enhance competitiveness, foster sustainable development, and ultimately improve citizens' quality of life.

Enhancing the effectiveness and efficiency of European Cohesion Policy investments requires rethinking how beneficiaries most in need of public support are identified. How can we know whether funded beneficiaries would have performed just as well without public funding? Or would other recipients have benefited more and generated even greater returns on public investments? Good targeting and well-tailored programmes are complex and require robust, reliable evidence on what works, for whom, under which conditions.

Improving the effectiveness and efficiency of European Cohesion Policy investments demands innovation in policymaking and experimentation to discover which investments work best for whom, under what conditions, and at what cost. To ensure continued relevance in today's rapidly evolving landscape, programmes must avoid static implementation approaches by ensuring access to and analysis of quality data as well as conducting rigorous analyses.[1] A commitment to continuous learning and adaptation is essential for maintaining a competitive edge and making informed public investment decisions. Experimental evaluations and policy experimentation provide valuable tools, through trial and error, to assess the effectiveness of implementation strategies, guide decision-making, and enhance accountability.

Enhancing the effectiveness and efficiency of European Cohesion Policy investments requires rethinking how evidence on what works is generated. Systematic reviews of Cohesion Policy evaluations emphasise the need to improve evaluation design and methods for more reliable findings, initiate data collection early, and invest in building and maintaining evaluation capacity. Despite extensive evaluations of Cohesion Policy instruments, the evaluation toolkit is still dominated by qualitative evaluations (incl. case-studies) or ex-post evaluations conducted long after investments are made, limiting opportunities to test, learn, and course-correct when it matters most (e.g., at mid-term). Recent, AI-powered analysis of 2.500 EU Cohesion Policy evaluations reveals major gaps that undermine the credibility of insights on public investment effectiveness.[2]

1 Policy experimentation allows for the rigorous measurement of the changes resulting from policies or interventions, for instance, focusing on changes in private sector outcomes such as business performance, managerial skills, and innovation practices.
2 For more information please refer to Asatryan et al. (2024).

A new strategic partnership supports the use of experimental evaluations –Randomised Controlled Trials (RCTs)–within Cohesion Policy, employing a method that can help identify what works and what does not, thus facilitating continuous improvement and adaptation to changing conditions. To foster creativity and promote the exploration of new policy ideas and implementation methods, the World Bank's Finance, Competitiveness, and Investment Global Practice (FCI GP), in collaboration with the European Commission's DG REGIO, launched a new initiative which aims to strengthen the monitoring and evaluation of innovation programmes in the EU. Since 2022, the partnership has sought to generate rigorous evidence in areas such as firm performance, productivity, entrepreneurship, innovation, and Small and Medium Enterprise (SME) development. By supporting the piloting of novel evaluation approaches, this partnership enables scalable innovations across the EU and beyond.

Total EU budget
(Multiannual Financial Framework)

€450 billion
Economic, social and territorial cohesion, including fisheries and rural communities

€218 billion
Less developed regions, with a requirement for Member States to allocate funds to regions

€20 billion
Facility for Union Actions

€10 billion
Interreg programme for territorial cooperation

€11,5 billion
support for Union Actions including cities, inter-regional innovation and cross-border projects

2028–2034
€2 trillion

Note: European Commission: Directorate-General for Internal Market, Industry, Entrepreneurship and SMEs, Europe's budget – Competitiveness – July 2025, Publications Office of the European Union, 2025, https://data.europa.eu/doi/10.2873/0084774

Short-term Causal Impacts of Innovative Investments into Competitiveness Policies

Policy experiments in Cohesion Policy are emerging as a means to both learn from and inform future policy cycles. Similar to how companies invest in R&D to identify the most successful products for scaling up, policy experimentation and rigorous evaluation can uncover novel interventions and the conditions under which returns on current investments are maximised. Moreover, random selection of beneficiaries among equally eligible candidates fosters fairness and transparency in the selection process. By ensuring all eligible firms have an equal chance of receiving funding, RCTs demonstrates that the selection process remains unbiased and free from favouritism. This, in turn, can enhance the credibility and legitimacy of the funding programme.

The first results from embedding experimentation within the Cohesion Funds are now available – they show that experimentation is not only feasible, but the results are also timely and insightful. This strategic engagement, titled *European Regional Competitiveness Policy Lab* ('EuroCOMP Lab'), followed an open call for Managing Authorities interested in participating. An initial workshop provided basic technical assistance for designing rigorous experiments, leading to the establishment of five innovative policy experiments in four Member States under Cohesion Policy.[3] This progress was made possible by substantial investments in building local and institutional capacities, alongside a strategic agenda that helped Managing Authorities navigate organisational challenges. Over the past years, the World Bank initiated the evaluations of the impact of instruments valued at over €100 million. These instruments were initially expected to launch in 2021/2022, with the new 2021-2027 programming cycle, but are still being rolled out in 2024.

3 Capacity-building events have helped cultivate a culture of evidence-based policymaking. Following an impact evaluation workshop in Lisbon in June 2022, which brought together 12 country teams from various Member States, six studies across five countries (Croatia, Czechia, Italy, Romania, and Spain) were selected for funding. For these projects, academic researchers, practitioners, and World Bank staff collaborated to develop rigorous evaluation designs. These experiments were tailored to the specific policies, knowledge gaps, and contexts, while also ensuring compliance with scientific standards.

Unlike before–after comparisons, RCTs achieve credible and unbiased comparisons between recipients and non-recipients of support, providing reliable estimates of investment impacts. Randomised Controlled Trials (**RCTs**) provide a robust method for evaluating interventions across various domains, offering insights into what works and under what conditions. Randomised Controlled Trials can help identify effective interventions by comparing outcomes with a comparison group, which serves as a *counterfactual* for what would have happened without the change (programme, approach, or investment).

RCTs are considered the gold standard for causal attribution in social sciences, as they eliminate confounding factors and selection biases. While observational studies rely on assumptions to replicate programme selection, RCTs ensure validity by randomly assigning treatment and comparison groups at the programme's initiation (hence the term *ex-ante*). This method eliminates selection bias, building a methodological framework for more credible evaluation results.

For example, in a business training programme, random assignment ensures that differences, such as firm size or location, are equally distributed between the trained and comparison groups, enabling the attribution of outcome differences to the training itself.

Short-term impacts of these policy evaluations, a few months after the investments were made, indicate that firms respond positively to the new policies and approaches. The primary questions addressed by the impact evaluations (IEs) seek to *better understand the impact of specific interventions on regional development*, examining how various approaches to policy implementation influence the results. Because eligibility rules, funding limits, and take-up risk differ across the five programmes, each one used a different experimental design suited to its context.

North-East Romania's digitalisation grants aim to boost SME productivity through targeted grants. The Cohesion Policy investment provides non-refundable grants to SMEs to support digital transformation. Using a randomised encouragement design, 519 out of 1.038 (most likely) eligible firms were selected to apply, with 333 eventually contracted. Early results show a 101% increase in digital investments, with notable rises in software, business administration, and production tools. While short-term impacts on profits and sales were not statistically significant, the programme shows promise for long-term digital growth.

Spain's R&D grant programme supports innovation in SMEs through a novel randomised evaluation. In Extremadura, a matching grant

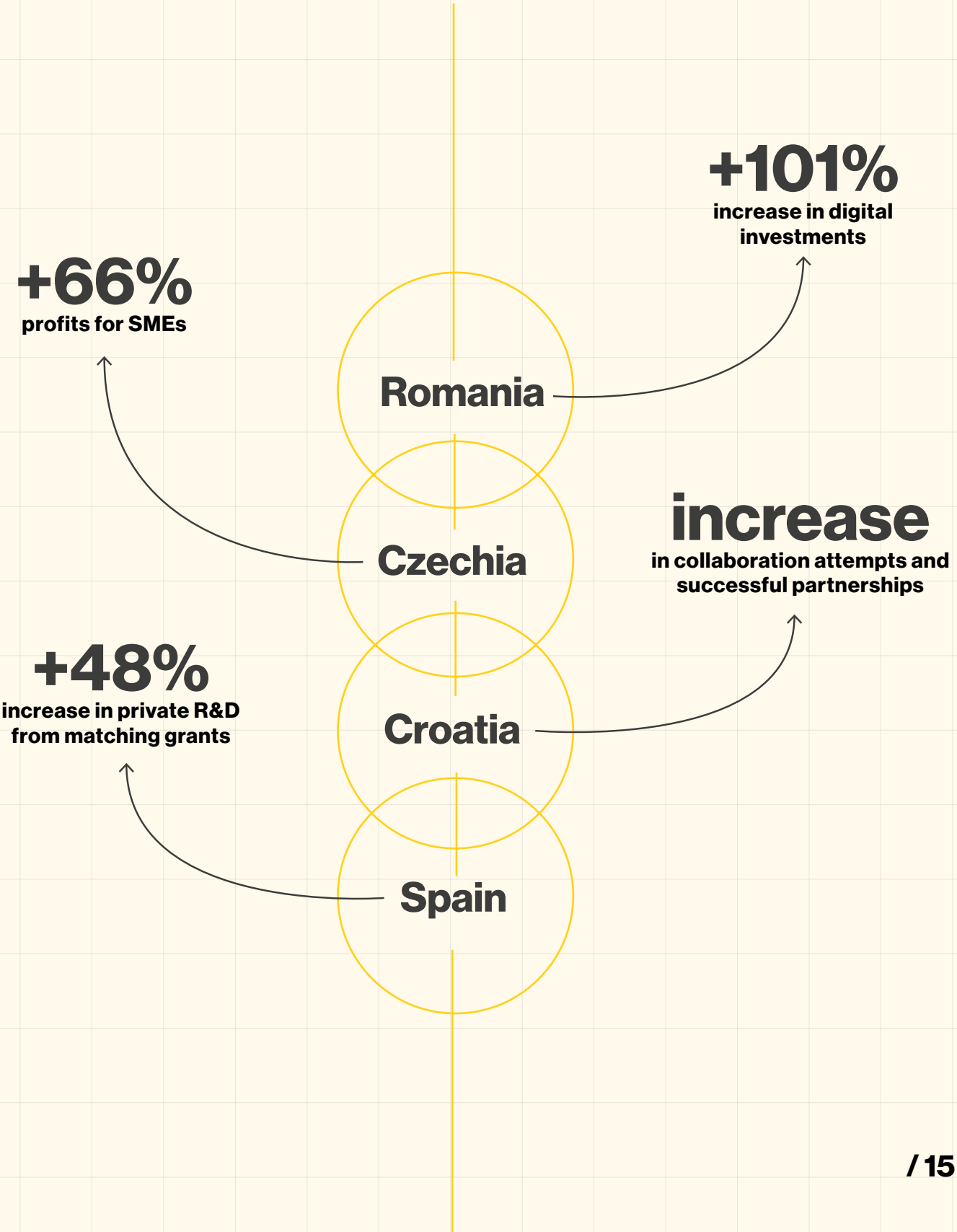
scheme offers up to €250.000 per firm to stimulate R&D, particularly in agriculture and ecological transition. A bin-ranking RCT design was used, where medium-quality applicants were randomly assigned to treatment and control groups. Supported firms projected a 48% increase in private R&D investment. The evaluation incorporated Bayesian methods and expert priors. The first short-term results also reveal that while public subsidies can significantly boost investment, they may also displace other funding sources.

Czechia's innovative training programmes enhance entrepreneurial and digital capabilities in post-coal regions. Two RCTs evaluated the impact of business and AI training under the Just Transition Programme. The Smart Start Programme in Ostrava improved business changes and monthly sales by 64%, emphasising scientific decision-making. The AI Klub in Usti and Karlovy Vary led to a 117% increase in monthly sales and a 66% rise in profits, along with significant gains in AI adoption and innovation. These results highlight the effectiveness of structured training and peer learning in revitalising disadvantaged regions.

Croatia's Scinergy pilot fosters science-business collaboration through structured matchmaking. The programme connects SMEs with scientists to support innovation, particularly in AI and energy efficiency. Using simple randomisation, 100 of 210 eligible firms were selected to participate in workshops and receive ongoing support. The initiative led to an increase in collaboration attempts and successful partnerships. AI-focused firms saw substantial gains in product development, commercialisation, and R&D employment. With the first results, the pilot already demonstrates high returns and supports scaling similar interventions.

Overall, this reliable, impartial, and credible evidence on the impacts of the evaluated programmes shall provide policymakers with the information needed to make decisions on resource allocation and improve outcomes for businesses and the broader economy. The collaboration has been designed to assist policymakers in identifying

what works, for whom, and under which conditions. By learning from short-term impacts of their investments, Member States can improve their understanding of regional markets and their challenges, enhance policy implementation processes, and ultimately design more impactful programmes.



Romania

"What is the impact of digitalisation support on Romanian SMEs' performance?"

Where	North-East region
Implemented by	Regional Development Agency (RDA)
Starting point	Romanian SMEs lag behind the EU average in digital performance; 67% of firms surveyed name lack of finance as their main obstacle
The programme	EU-funded grants of EUR 15.000–100.000 for digital tools, equipment, and services
Who is randomised	1.038 eligible SMEs (pre-assessed on digital-performance indicators) from over 1.500 applicants
Design	Randomisation among pre-identified applicants (with encouragement to apply): 519 firms randomly selected and encouraged to formally apply vs. 519 not selected (comparison group) • Strengths: requires (and thus eventually ensures) wide outreach due to simple application process • Limitations: firms may drop out after application, affecting statistical power • Requirements: large sample size and encouragement to apply are required; simplified application and contracting process
Process	Pre-assessment survey → Random selection → Encouraged to apply → Feasibility study (76%) → Firms contracted (64%)
Outcomes tracked	Digital investments, firm performance

Spain

"What is the impact of R&D grants on SMEs' business performance?"

Where	Extremadura
Implemented by	Regional Government of Extremadura (JUNTA) with the Foundation for the Development of Science and Technology and the Extremadura Science and Technology Park Foundation (FUNDECYT-PCTEx)
Starting point	Market failures keep regional R&D investment below its optimal level
The programme	Matching grants of up to EUR 250.000 per firm (average EUR 138.000) for R&D projects
Who is randomised	SME applicants (mainly agriculture and ecological transition), scored on 11 criteria
Design	Bin-ranking randomised controlled trial among intermediate-range firms (22 funded vs. 34 not funded) • Strengths: leverages investment return uncertainty for applications with unclear returns • Limitations: requires a large sample size in the middle range (bin) • Requirements: clear, replicable criteria to determine groupings (bins)
Process	Applications → External scoring → Middle-bin lottery → R&D grants → Follow-up survey
Outcomes tracked	Private R&D investment, implementation success, expected sales growth

Czechia — Smart Start and AI Klub

"Does light-touch support to SMEs improve firm capabilities and performance?"

Where	Smart Start: Ostrava, Moravian-Silesian region; AI Klub: Ústí and Karlovy Vary regions
Implemented by	Just Transition Programme
Starting point	Entrepreneurial activity remains low in regions hit by the coal industry's decline
The programme	Smart Start: Seven workshops on scientific decision-making and customer validation; AI Klub: Five hands-on workshops on AI for efficiency, marketing, sales, and finance
Who is randomised	Entrepreneurial voucher applicants are randomly divided into two groups. The comparison group receives just the vouchers, the treatment group receives additional support.
Design	Simple lottery, unequal shares. Smart Start: a "voucher plus support" group (163 firms), which also receives the Smart Start training, and a "status quo" group, which receives only the standard vouchers (83 firms). AI Klub: group invited to AI Klub's networking and support workshop (117 firms) vs. 56 firms that only received vouchers. • Strengths: provides detailed comparisons across multiple interventions • Limitations: managing multiple treatments (program variations) can be complex • Requirements: large sample size of firms interested in add-ons; effective outreach campaign to promote extra support; strategy to reduce contamination between treatment arms
Process	Interest sign-up → Random selection → Seven workshops → Presentations → Follow-up survey
Outcomes tracked	AI adoption, innovation, firm performance

Croatia

"Does connecting firms with scientists foster research partnerships and innovation?"

Where	Croatia
Implemented by	Innovation Centre Nikola Tesla with the Ministry of Science, Education, and Youth
Starting point	Only 8% of Croatian firms collaborate with higher education, far below the EU average
The programme	Matchmaking events, workshops, and ongoing support for firm-scientist pairs
Who is randomised	210 eligible firms (from over 250 applications) seeking scientific collaboration
Design	Simple lottery among applicants: 100 firms invited to Scinergy vs. 110 not invited • Strengths: requires (and thus eventually ensures) wide outreach due to simple application process • Limitations: firms may drop out after application, affecting statistical power • Requirements: eligibility criteria should be clear and quick to evaluate (e.g., no in-depth legal background checks). Commitment to participate should be ensured before randomisation (e.g., through signed expressions of interest)
Process	Applications → Lottery → Matchmaking event → Joint R&D planning → Follow-up survey
Outcomes tracked	v

NOTE

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Romania: Financial Support to Digitalise

“What is the impact of digitalisation on Romanian firms’ performance?

Investment

Romania's programme provides non-refundable grants to SMEs for digitalisation, aiming to boost productivity in the North-East region, with a focus on key digital performance indicators. Digitalisation has the potential to transform industries, particularly SMEs, which in Romania lag behind the EU average in digital performance. Key barriers to digital transformation include a lack of financing, awareness, and capacity to capitalise on digital investments. In North-East Romania, digitalisation grants are awarded to eligible SMEs under the de minimis aid scheme, with a total allocation of €31.000.000 in the first call (€26.350.000 ERDF and €4.650.000 from the state budget).[4] Firms can receive grants ranging from €15.000 to €100.000, covering 90% of eligible project costs. The World Bank studies the impact of digitalisation on SME performance. The hypothesis is that financial support will boost productivity by enabling the acquisition of knowledge, services, and soft/hardware. As part of the study, the World Bank conducted a pre-assessment using Digital Economy and Society Index (DESI) indicators, as defined by the Regional Development Agency (RDA): The most common technology was "20+ percent of employees using connected mobile devices" (681 businesses, 66% of the sample), followed by "Internet connection of at least 30 Megabits per second (Mbps)" (620 businesses, 60%) and "50+ percent of employees with Internet access" (604 businesses, 58%).[5]

4 Call guidelines section 3.3, p.11.
5 Table 1.1 lists the technologies included in the indicator and the percentage of businesses using them.

Table 1.1: Share of Firms Meeting Each DESI Indicator in North-East Romania

DESI Indicators	Number of Firms	% of Firms
20+ percent of employed people use connected mobile devices	681	65,6
Internet connection of at least 30 Mbps	620	59,7
50+ percent of employed people w/ Internet access	604	58,2
Use a social media network	546	52,6
Own a website or a web page	516	49,7
Issue electronic invoices that allow automatic processing	395	38,1
Smartphones	373	35,9
Pay for internet advertising	279	26,9
It has a website with complex functions	130	12,5
IT uses Cloud Computing, high or medium complexity	108	10,4
Sales from e-Commerce activity exceed 1 percent of total turnover	105	10,1
Business-to-consumer web sales exceed 10 percent of total online sales	101	9,7
None of the above	43	4,1

Note: Each row is one of the 12 individual components (binary technology-adoption indicators) as defined by the North-East Regional Development Agency (RDA) making up the Digital Economy and Society Index (DESI) score, which sums these on a 0–12 scale. N = 1,038 eligible North-East Romanian MSMEs (identified June–Oct 2023 as part of an EU funding application process). "Number of Firms" = firms meeting that component; "% of Sample" = corresponding share. Firms can meet more than one component.

Experiment

In an experimental encouragement design, a randomly selected subset of participants is incentivised to join a programme. This allows evaluation by comparing outcomes between encouraged and non-encouraged groups, adjusting for actual participation rates.[6] Designing a compelling encouragement is crucial for the success of this approach. In collaboration with the North-East RDA, the World Bank established a process to ensure compliance with legal regulations for evaluating and

6 For example, among 2.000 pre-selected startups eligible for a training programme, 1.000 are randomly encouraged (via financial incentives or expert support) to participate. At the same time, the other 1.000 receive no encouragement to enroll in the programme (despite being eligible). After one year, business success indicators (e.g., business survival, revenue growth) are compared between both groups, accounting for actual participation. In this evaluation approach, the effectiveness of the encouragement to generate significant take-up (in the so-called "first stage") is key. However, if few or none take up the programme, it suggests ineffective encouragement, making evaluation difficult and potentially underestimating the programme's impact. Low take-up of the programme after encouragement prevents any meaningful review of the programme's impact since the treatment and comparison groups would effectively be the same in terms of programme participation.

implementing the digitalisation programme. The following steps were designed to **integrate the RCT** and derive key insights on improving digitalisation in the region:

1 First, the RDA and World Bank formed a learning partnership, publicly announced at the programme's launch. The first call for proposals was defined as non-competitive and framed as a pilot to generate learning. Beyond implementing the RCT, the goal was to adapt future digitalisation programmes better to meet the needs of SMEs. To this end, SMEs participated in a *pre-assessment survey* to (a) assess digital maturity, (b) express interest in applying for the programme, and (c) receive an automated, simplified preliminary eligibility assessment.[7] Developed in Spring 2023 and piloted in May, the pre-assessment concluded in September 2023, identifying 1.038 likely eligible SMEs from over 1.500 applicants.

2 Second, in October 2023, 519 firms were randomly selected from the eligible pool of 1.038 for funding. This random selection ensured comparability between selected and non-selected groups across various characteristics. The selection, conducted in front of a notary (Figure 1.1), ensured transparency, with the remaining 519 firms serving as a comparison group.

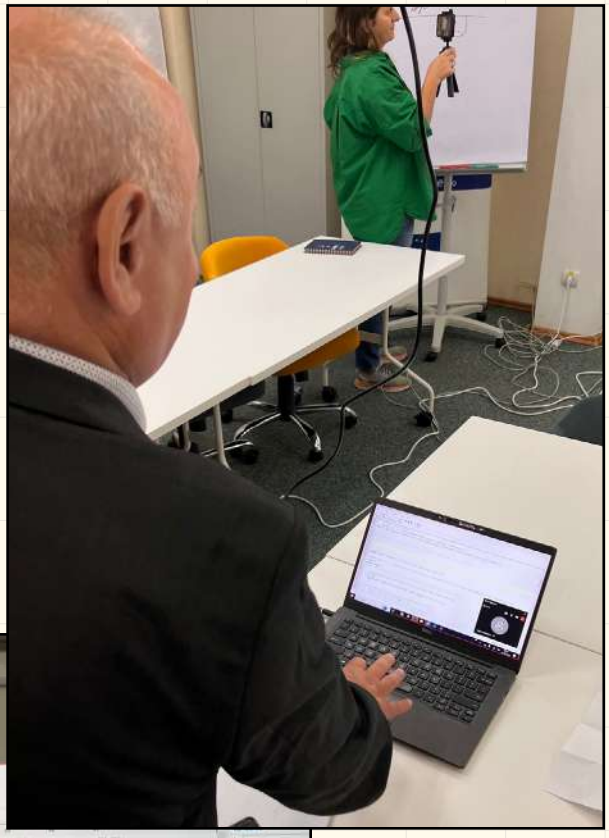
3 Third, the selected 519 firms were invited to submit detailed digitalisation investment proposals via the national MySMIS IT platform. Of those, 372 firms (71,7%) submitted final documents, including the required digital feasibility study designed by RDA by the end of Winter 2024. After verifying eligibility, RDA is finalising contracts.

7 The survey checked eligibility using objective data, cross-verified with administrative information to streamline RDA's review process.

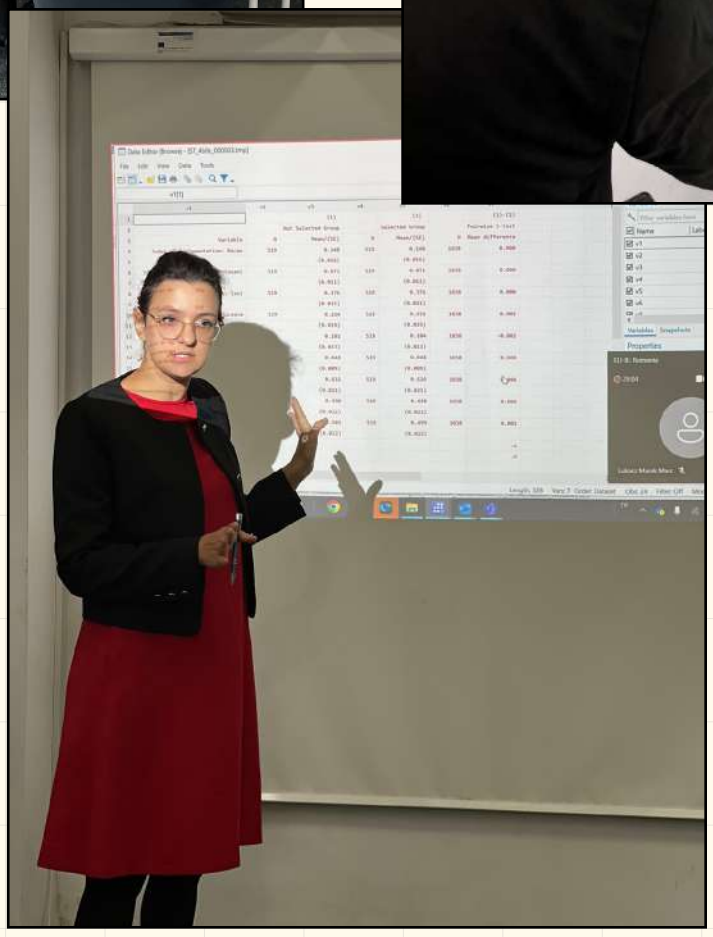
Figure 1.1: Randomisation Process with Notary's Presence in Romania



(a) Drawing the Random Number Seed



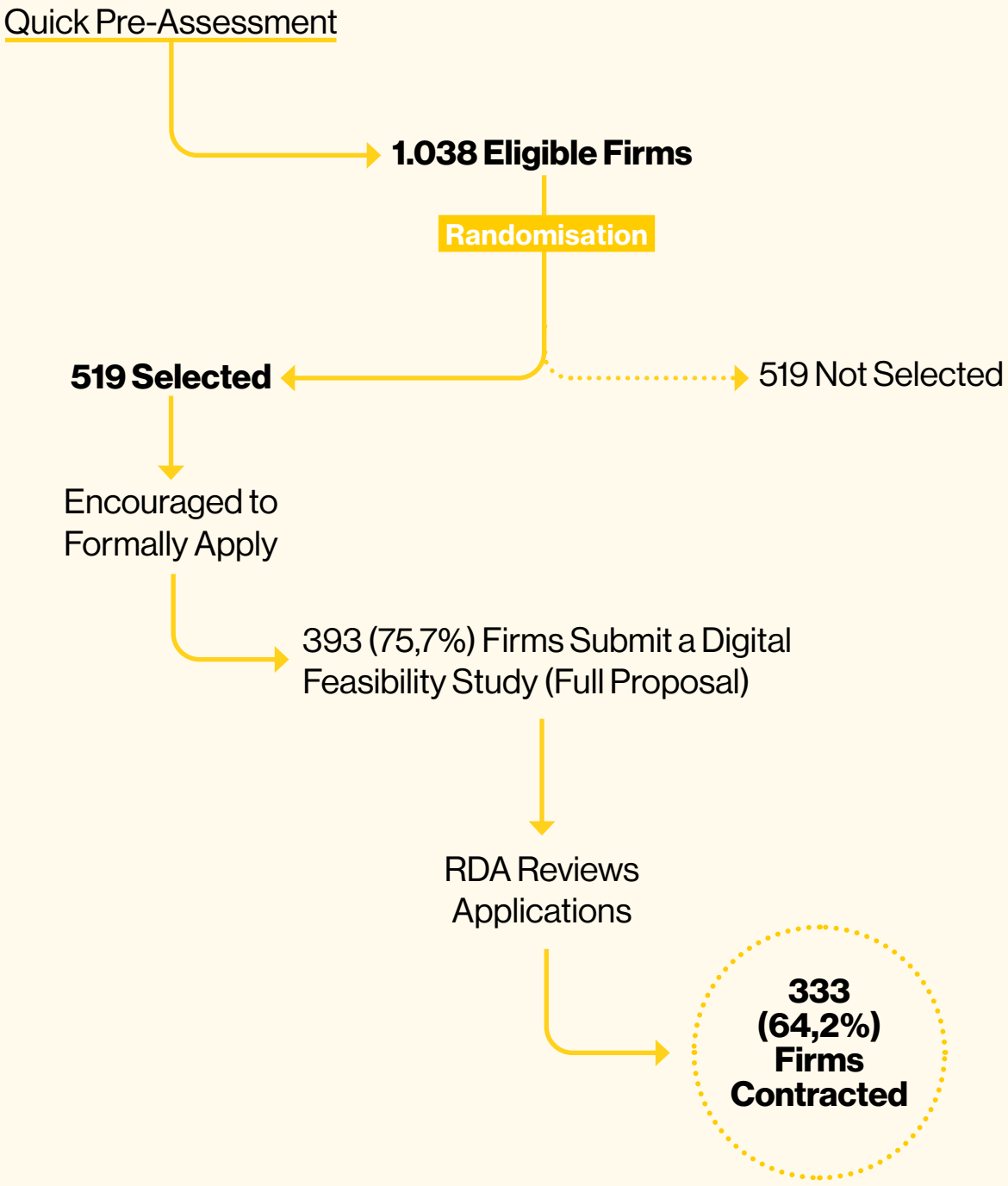
(b) Running the Code to Select the Firms after Inserting the Seed



(c) Reviewing the Balance Statistics

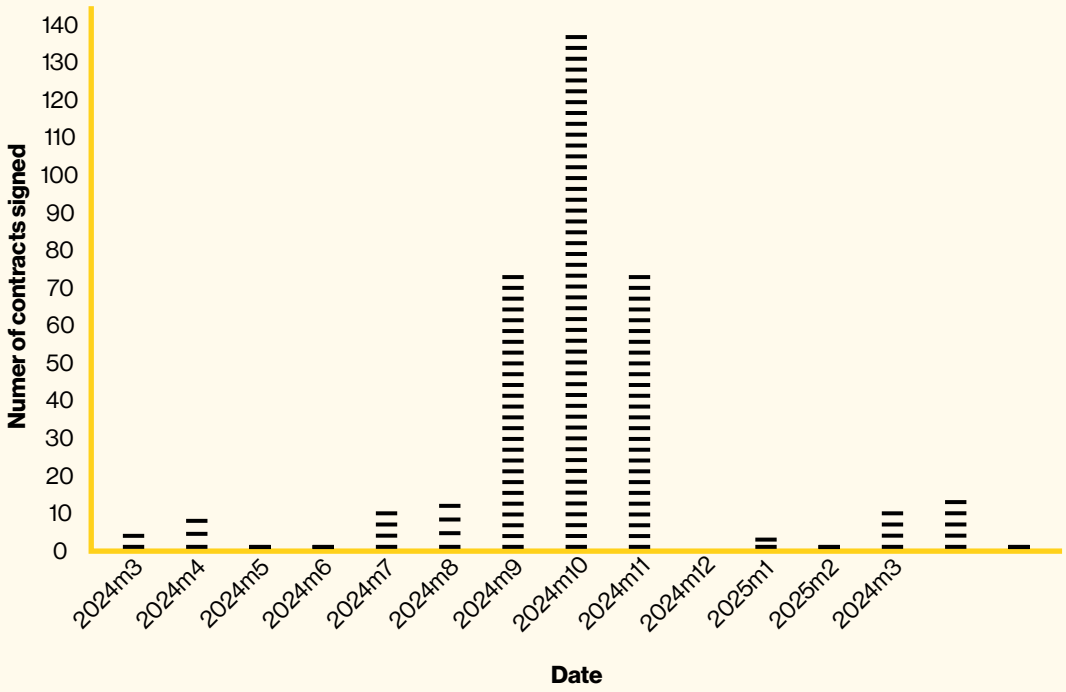
Source: Randomisation event at the Regional Development Agency, October 2023.

Figure 1.2: Experimental Design in Romania



Note: Breakdown of the application process stages and experimental design. Only 76% of selected firms chose to formally apply for funding, and only 64% of selected firms were contracted as of 31 March 2025.
Source: Implementation data from RDA.

Figure 1.3: Contract Signatures, by Calendar Month



Note: Number of contracts signed between March 2024 and March 2025. Terminated contracts are not included.
Source: Implementation data from RDA.

Short-Term Impacts

Digital investments increased by 101%. The primary evaluation metric is the Intention-to-Treat (ITT) effect, capturing the average impact of being offered the opportunity to apply for funding (as opposed to actually receiving the funds).[8] Treated firms invested an average of RON 71.199 in digital tools, compared to a control group mean of RON 70.529, representing a 101% increase relative to the control mean.

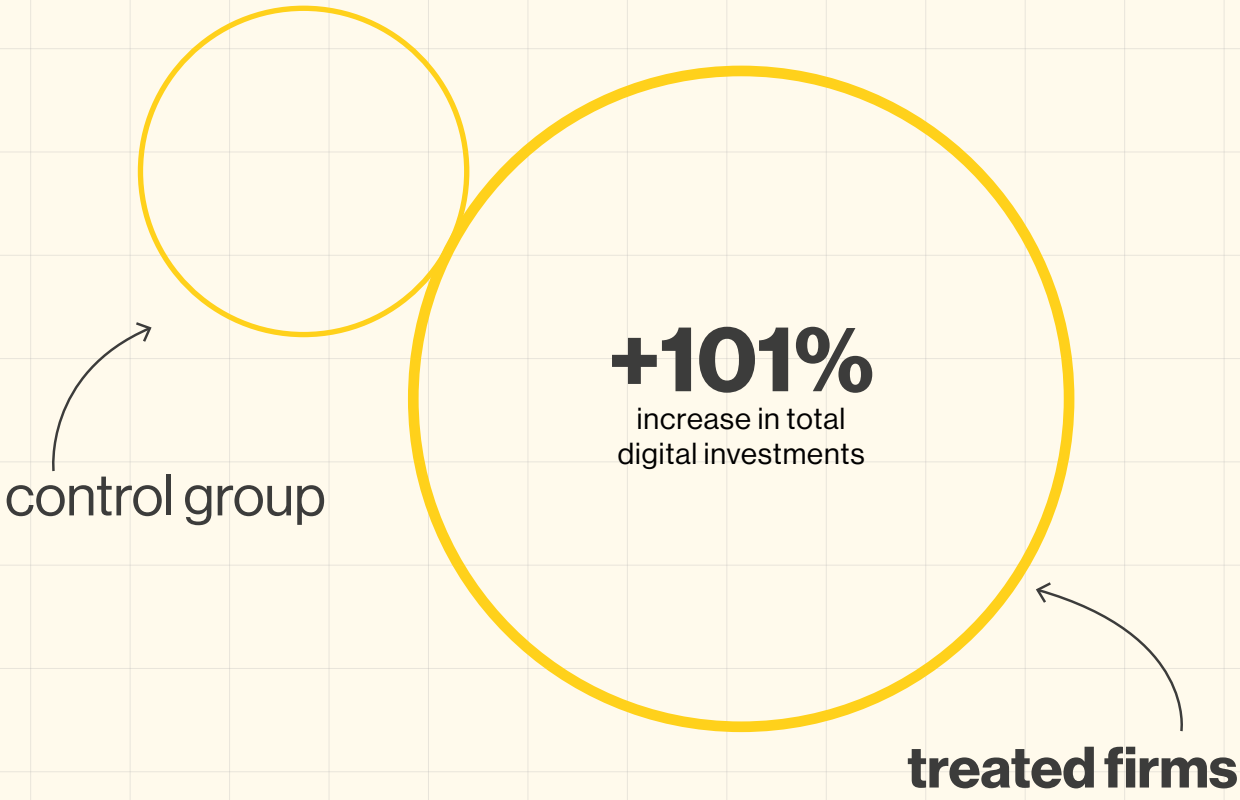
8 The impacts of receiving the funds are reported in the full report.

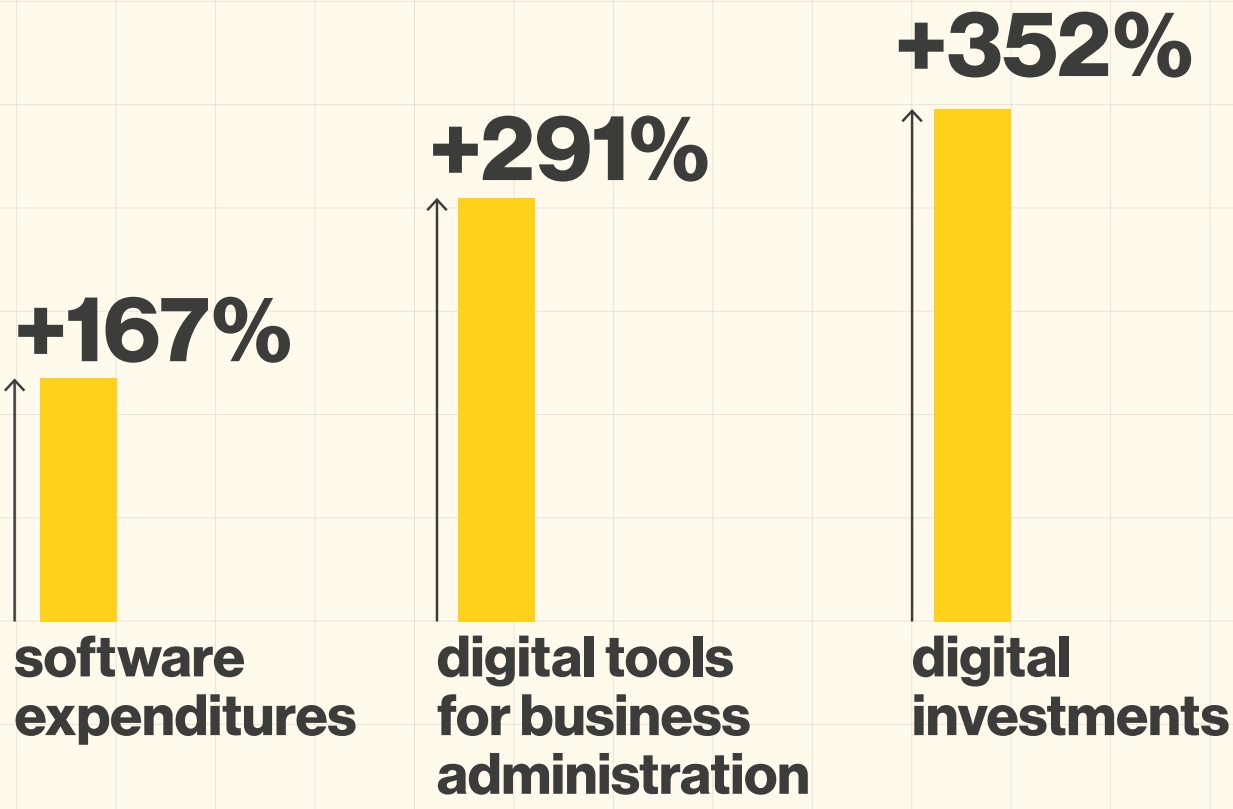
Software expenditures rose by 167%, digital tools for business administration saw a 291% increase, and production-related digital investments increased by 352%. Results are reported for all firms, independent of whether they eventually signed the grant contract with RDA or not, and for those who did so, they are reported only a few months after contract signature. In other words, the reported estimates are very conservative estimates of the treatment effects based on self-reported survey data. Firms in the treatment group spent RON 8.696 on software, compared to RON 5.207 in the control group, a 167% increase. Moreover, expenditures in business administration tools rose to RON 37.082 among treated firms, compared to RON 12.752 in the control group. Treated firms spent RON 40.380 on production tools, compared to RON 11.474 in the control group.

Furthermore, marketing tool investments rose by 118% and sales tool investments increased by 241%. Marketing-related digital expenditures increased to RON 209 among treated firms, compared to RON 177 in the control group. Treated firms also spent RON 3.312 on sales tools, compared to RON 1.374 in the control group.

Customer Relationship Management (CRM) usage increased by 26% and Enterprise Resource Planning (ERP) usage increased by 22%. The share of firms using CRM software rose by 3,1 percentage points from a control mean of 11,7%. ERP adoption rose by 5,6 percentage points from a control mean of 26%.

On average, no significant short-term impact on profits or sales was observed yet. While treated firms showed slightly higher sales and labor productivity, the differences were not statistically significant, likely due to the early stage of implementation. Treated firms were 10,7 percentage points more likely to attribute future revenue growth to new customers, compared to a control mean of 44%. Overall, while performance outcomes are not yet evident, the early results suggest strong potential for long-term impact. Tailored support and continued monitoring will be essential to ensure inclusive and sustained digital transformation.


control group
+101%
increase in total
digital investments
treated firms


+167%
software
expenditures
+291%
digital tools
for business
administration
+352%
digital
investments

NOTE

2.



Spain: Financial Support to Innovate

“What is the impact of R&D grants on SMEs business performance inWestern Spain?

Investment

Spain's matching grant programme awards up to €250.000 per firm to boost R&D investment in small and medium companies, primarily in agriculture and ecological transition. To address market failures that keep regional R&D investment at suboptimal levels, the Junta de Extremadura, the Regional Government of Extremadura (JUNTA), and the Foundation for the Development of Science and Technology and the Extremadura Science and Technology Park Foundation (FUNDECYT-PCTEx) have developed a matching grant programme funded by the EU, offering large R&D grants to small and medium firms in Extremadura. Each cycle plans to award €3,5 million in grants (up to €250.000 per company), financing 35% to 80% of proposed R&D investments.

Applicants are primarily small and medium companies, mainly in the agricultural or ecological transition sectors, with most grants supporting industrial research and experimental development. The study evaluates the impact of large public R&D grants in Extremadura, Spain, using a randomised controlled trial (RCT) – the first of its kind for such high-value subsidies (average approximately €138.000). The subsidy programme, funded by the EU and implemented by the Junta de Extremadura and FUNDECYT-PCTEx, supports R&D in agro-industry, ecological transition, health, digital transformation, and tourism.

Experiment

In this RCT design, potential beneficiaries (e.g., firms) are ranked based on their application scores during the evaluation process, which considers their performance and eligibility.[9] In Spain, the World Bank is collaborating with JUNTA to offer technical support aimed at generating valuable insights from this investment. A rigorous impact evaluation requires a sufficient number of high-quality applications. To achieve this, the World Bank collaborated with the JUNTA and FUNDECYT-PCTEx to coordinate an advertising campaign aimed at attracting at least 100 beneficiaries.

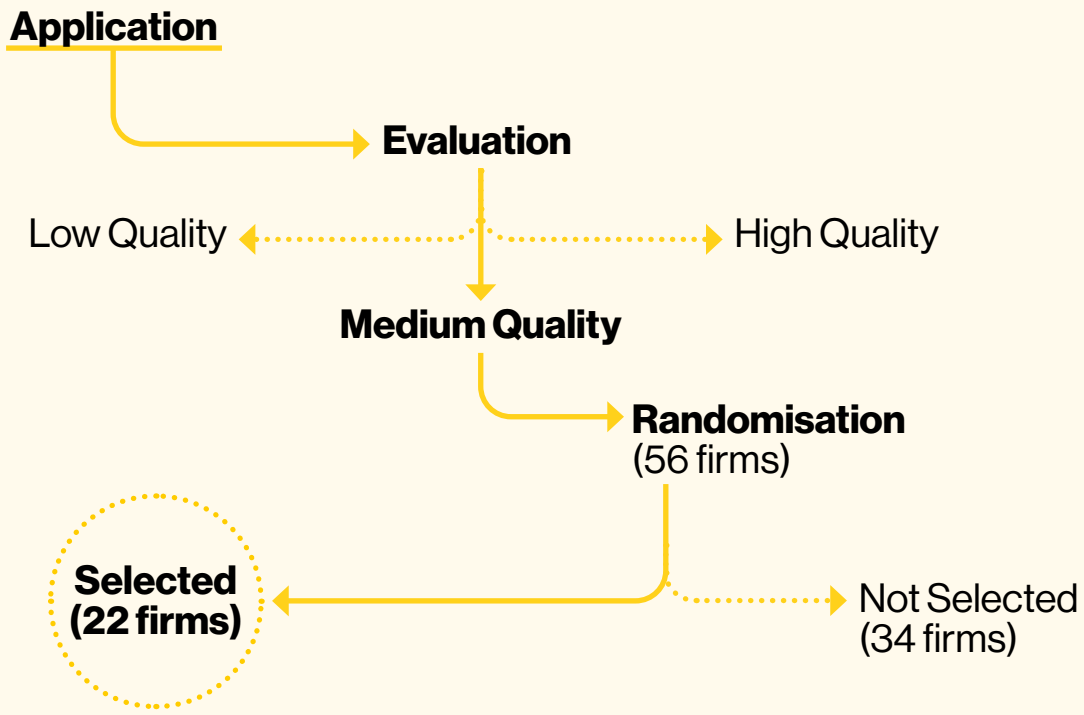
All applications were evaluated by external reviewers following pre-established guidelines and classified into three groups based on quality: group 1 (poor-quality proposals) that do not meet minimum standards, group 2 (intermediate-quality proposals) in the "intermediate" range, and group 3 (highest-quality projects) that are directly selected to receive grants. In more detail, proposals were scored on 11 objective criteria and categorised into high, medium, and low potential. High-potential proposals were automatically funded; low-potential were rejected. Medium-potential proposals (56 total) were randomly assigned to treatment (22 funded) and control (34 not funded) groups.[10] Randomisation successfully balanced key firm characteristics, and there was no evidence of selective attrition[11] in follow-up survey responses.

9 In more detail, applicants are sorted into bins according to their scores, with the highest and lowest performers excluded from the evaluation study. High scorers are automatically accepted into the programme, while low scorers are rejected. The middle bin of applicants, equally eligible for the programme, is randomly assigned either to the treatment (programme/funding) or comparison group. This design is effective when there are clear eligibility criteria and resources are limited. It ensures that top performers are prioritised while still providing a randomised comparison group to assess the programme's impact among participants with ex-ante unclear or disputed gains. For an example of this design in practice, see Karlan and Zinman (2011).
10 Randomisation was based on budgeted investment size to ensure balance across firm sizes and project scales.
11 Firms that dropped out of the follow-up survey were not systematically different between the funded and non-funded groups.

Short-Term Impacts

Firms receiving the subsidy projected €65.000 more in private R&D investment for 2025 – a 48% increase over the control group average. Data collection included a baseline survey (pre-randomisation), a follow-up survey (three months post-randomisation), expert priors, and risk, ambiguity, and time preference surveys for evaluators. Additionally, the evaluation incorporated expert priors to improve precision and account for high variance and small sample size. The team analysed new Bayesian methods as a promising approach for evaluating complex, high-variance innovation programmes. Before programme implementation, expert priors revealed, for instance, that without funding, the average probability of successful implementation was 16%, which increased to 67% with funding. Expected sales growth (conditional on success) doubled from 8% to 16%.

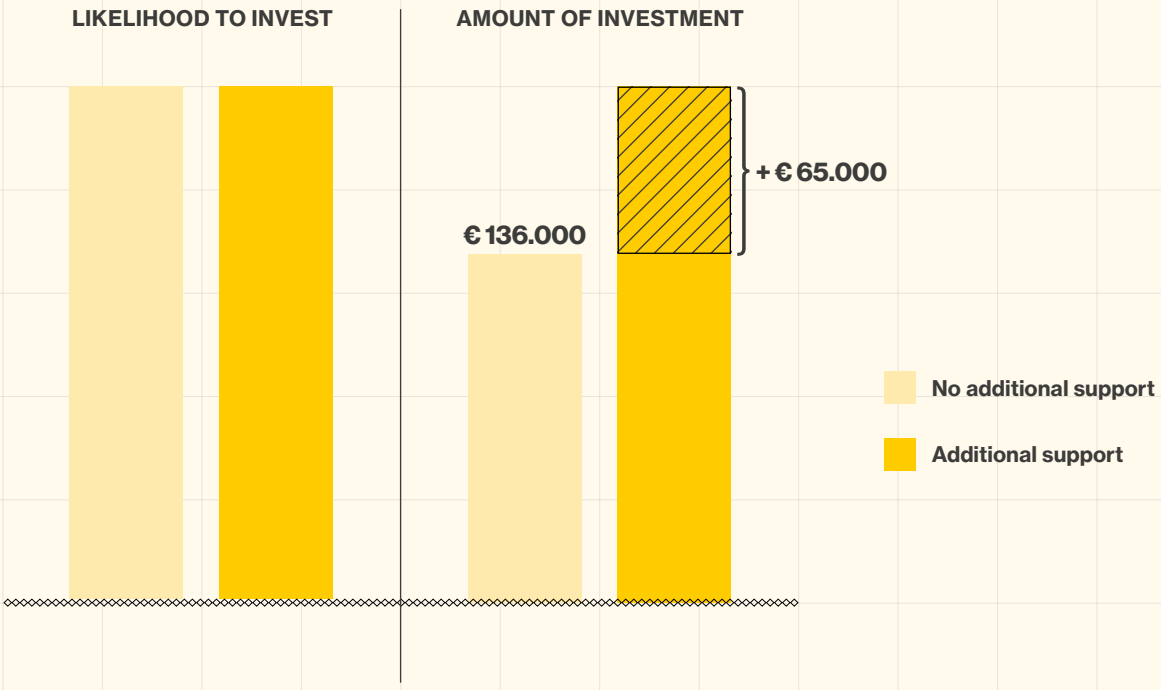
Figure 2.1: Experimental Design in Spain



Treated firms exhibited greater growth in their annual R&D investments relative to the baseline period. The results show a sharp rightward shift in the distribution of R&D growth among treated firms, measured in percentage terms. At the left tail are firms in the control group that were previously investing in R&D but now have no ongoing projects, leading to a growth of -100%. At the right tail are treated firms that previously invested minimal amounts but are now commencing large-scale R&D projects.

Overall, public R&D subsidies can significantly boost private investment, but careful targeting is needed to avoid displacing other funding sources. Over half of the control group firms planned to continue their R&D projects using alternative funding or internal resources, suggesting that public subsidies may crowd out other funding sources.

Projected R&D investment for 2025



Note: European Commission: Directorate-General for Internal Market, Industry, Entrepreneurship and SMEs, Europe's budget – Competitiveness – July 2025, Publications Office of the European Union, 2025, https://data.europa.eu/doi/10.2873/0084774

NOTE

3.



Czechia: Identifying the Right Training and Additional Support for Firms

“Does light-touch support to SMEs improve firm capabilities and performance?

Investment

In Czechia, the World Bank evaluates whether enhancing entrepreneurial skills improves the effectiveness of a voucher scheme supporting business development, innovation, and digital transformation in regions affected by the coal industry's decline. Several Czech regions are focused on economic diversification, prioritising private sector growth, but entrepreneurial activity remains low. A new programme, approved by the European Commission in September 2022, targets regions affected by the decline of the coal mining industry. It emphasises business R&D, post-mining area regeneration, skills development, clean energy, and the circular economy, with a total allocation of €1,641 billion. A key component of the Just Transition Programme (JTP) is the voucher scheme for entrepreneurs in Czech coal regions, offering different types of financial support.[12] These vouchers are available in the Moravian-Silesian, Ústí, and Karlovy Vary regions, with a focus on diversifying local economies reliant on carbon-intensive industries. As part of two independent impact evaluations, additional activities were tested to improve business outcomes in these regions, partially also targeting innovation voucher recipients. For this, the World Bank has been working with local partners and chambers of commerce to extend the additional support to new potential beneficiaries.

12 The first is the Voucher for Business Development, which provides 50.000 CZK (€2.100) for new entrepreneurs and up to 0,5 million CZK (€21.100) for SMEs to start or expand businesses, improve processes, or receive coaching and education. The second is the Innovation Voucher, offering SMEs up to 1 million CZK (€42.200) to develop new products, optimise services, enhance processes, and create innovation-related jobs. The third is the Digital Voucher, granting up to 0,5 million CZK (€21.100) to assist SMEs in preparing for digital transformation by acquiring hardware and software.

3.1 ScientificDecision-Making for Entrepreneurial Growth in Ostrava

Training

In the Moravian-Silesian region, the pilot aimed to foster entrepreneurial growth by addressing two key constraints: limited ability to validate business models and low ambition to scale. In particular, the programme emphasises *mindset training* for innovation, offering *"smart training"* that promotes a scientific approach to decision-making under uncertainty. This training, aimed at firms selected for entrepreneurial vouchers (and others), teaches participants to make informed decisions in uncertain environments, covering topics such as understanding innovation, risk versus uncertainty, hypothesis testing, and A/B testing, culminating in participant presentations (Figure 3.1).

Experiment

Firms selected for entrepreneurial vouchers are randomly divided into two groups: a "voucher plus support" group, which also receives the Smart Start training, and a "status quo" group, which receives only the standard vouchers. The RCT involved 246 firms, with 163 assigned to the treatment group ("voucher plus support" group) and 83 to the control group ("status quo" group). The Smart Start programme, implemented in partnership with the Moravian Silesian Innovation Centre (MSIC), delivered seven workshops focused on scientific decision-making, customer validation, and structured business development. The evaluation measured business changes, motivations for change, and firm performance.

Short-Term Impacts

The short-term results indicate that the Smart Start programme significantly improved externally-oriented business changes and revenue growth among early-stage entrepreneurs in Ostrava. The scientific decision-making framework helped participants adopt a more structured, proactive approach to customer targeting and sales.

Figure 3.1: Smart Start Training in Czechia



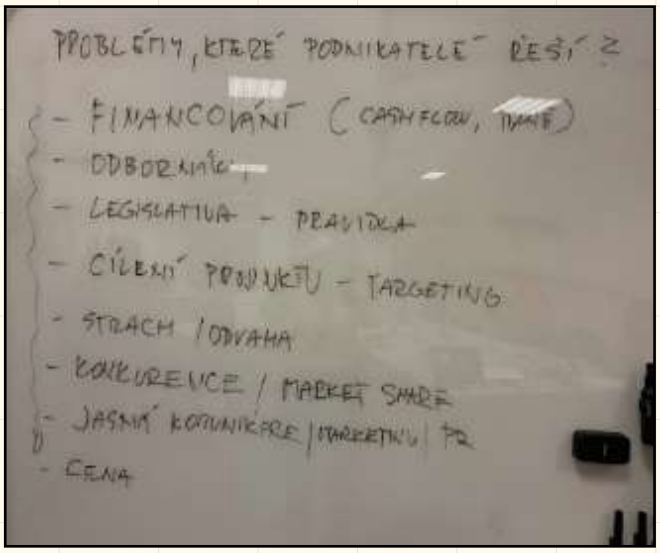
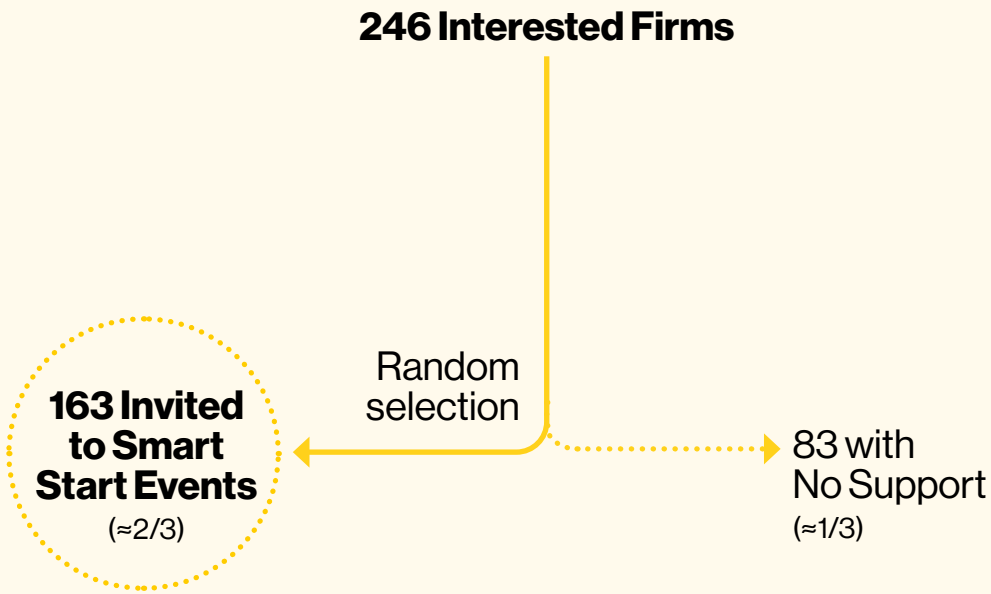






Figure 3.2: Experimental Design for Smart Start Training in Czechia



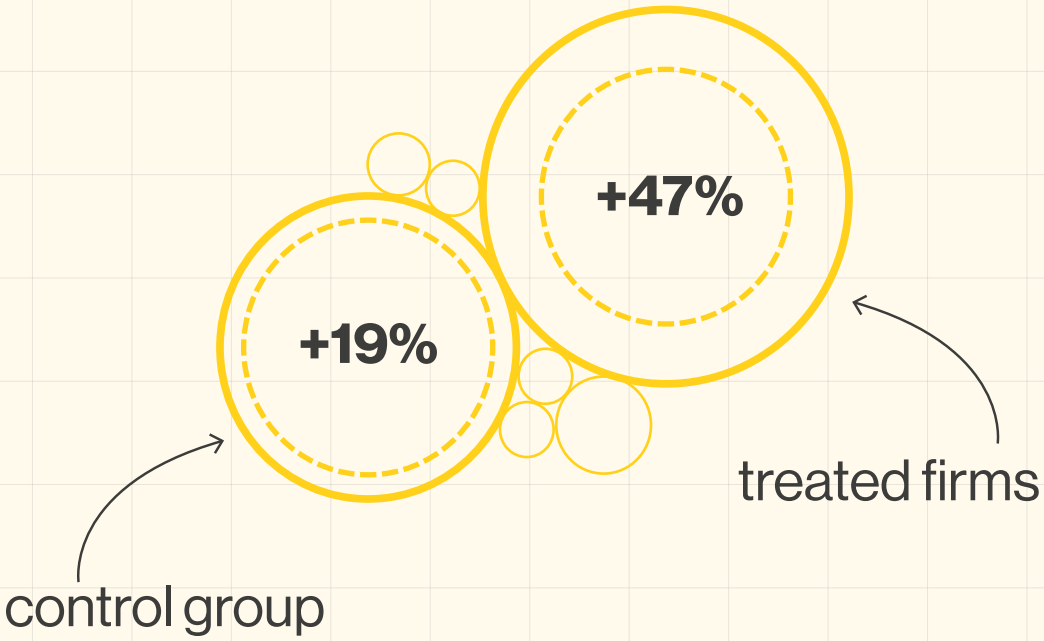
In more detail, the Go-to-Market (GTM) change intensity increased by 47%. Treated firms scored 0,65 points higher on a 1–7 scale than the control mean of 1,39. 42% of treated firms made moderate or greater GTM changes, compared to 19% of control firms. Moreover, treated firms implemented 12% more high-intensity GTM changes than the control mean of 0,25.

For Production and Operations Management (POM) changes, the differences were not significant. Treated firms scored only 0,17 points higher than the control mean of 1,58 (11% increase, not significant).

Programme-related motivations for change increased by 98%. Treated firms cited 2,10 programme-related reasons on average, compared to 1,06 in the control group. Treated firms were also 24 percentage points more likely to be in the top half of programme-related motivations (29% control mean).

Monthly sales increased by 64%. Treated firms reported CZK 103.325 higher monthly sales than the control mean of CZK 160.271. Overall, the sales index improved by 0,32 standard deviations.

Go To Market changes



3.2 AI Adoption and Innovation in Post-Coal Region

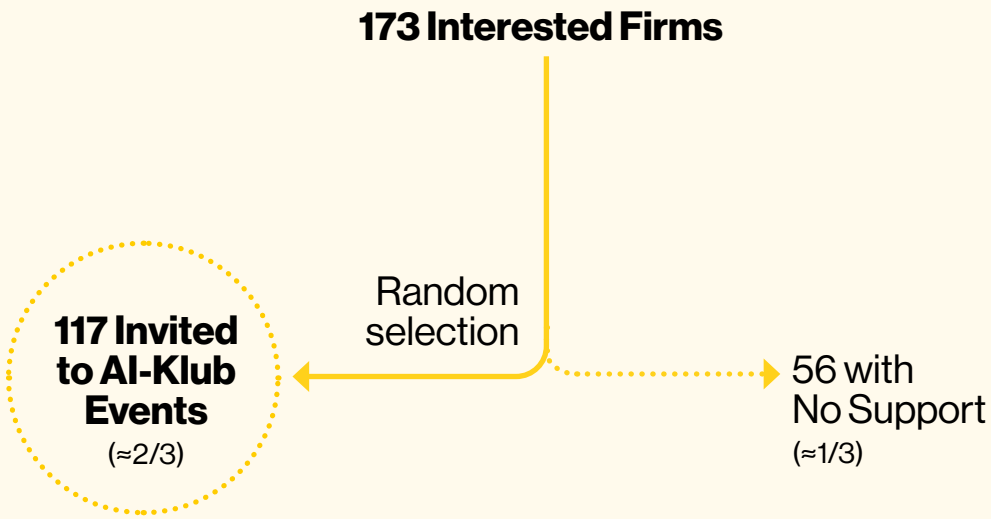
Training

The AI Klub programme offered five in-person workshops covering AI applications in business efficiency, marketing, sales, product development, and financial management. The impact evaluation includes complementary activities to strengthen the JTP outcomes. One such initiative is the *'Smart Start'* project, launched with World Bank's support since 2022. In the ˇUstí and Karlovy Vary regions, this programme focuses on innovation vouchers, incorporating a *'networking'* approach, where firms receive expert advisory support and are encouraged to share knowledge and network to enhance innovation. The workshops emphasised hands-on learning, peer-to-peer exchange, and practical application of AI tools.

Experiment

As in Ostrava, the comparison is between a "voucher plus support" group – firms that are also invited to the AI Klub's networking and workshop support – and a "status quo" group that receives only the standard vouchers. To form the eligible pool, all voucher recipients were invited to express interest in the support programme, as well as firms outside of this pool. Approximately ⅔ of interested firms were invited to participate in the events. The unequal allocation (with more firms in the treatment group) allows for the testing of further add-on activities in future. The RCT involved 173 firms, with 117 assigned to the treatment group and 56 to the control group. The evaluation measured AI adoption, perceived effectiveness, innovation, business networking, and firm performance.

Figure 3.3: Experimental Design for AI Training in Czechia



Short-Term Impacts

The AI Klub programme significantly boosted AI adoption, innovation, and firm performance in structurally disadvantaged Czech regions. The group-based, hands-on workshop model proved effective in overcoming barriers to technology adoption and fostering peer learning. These findings support scaling similar interventions to accelerate digital transformation and economic revitalisation in post-industrial regions.

In more detail, AI adoption intensity increased by 15% and AI use in customer management rose by 118%. Treated firms scored 0,38 points higher on a 0–5 AI adoption scale, compared to a control mean of 2,62. Treatment led to a 0,72-point increase in AI use for customer management over a control mean of 0,61.

AI use in product development increased by 48% and AI use in marketing rose by 31%. Treated firms scored 0,48 points higher than the control mean of 1,00. Treatment firms scored 0,58 points higher than the control mean of 1,87.

Perceived AI effectiveness increased by 43%. Treated firms reported a 0,68-point increase on a 0–5 scale, relative to a control mean of 1,59. Perceived AI effectiveness also rose in marketing and customer management by 71%. Both functions saw 0,52 and 0,47-point increases, respectively, over control means of 0,73 and 0,66.

Individual-level innovation increased by 67% and firm-internal innovation rose by 55%. Treated firms scored 1,04 points higher than the control mean of 1,56. Treatment led to a 0,62-point increase over a control mean of 1,13.

New customer acquisition increased by 60%, monthly sales increased by 117%, and monthly profits increased by 66%. Treated firms scored 0,65 points higher than the control mean of 1,09. Treated firms saw a CZK 1.420.659 increase over a control mean of CZK 1.218.750. Treatment also led to a CZK 72.648 increase over a control mean of CZK 110.484. Overall, a composite measure reflects consistent gains across sales and profits. Firm growth index improved by 0,78 standard deviations.

New business connections increased by 78%. Treated firms reported 3,82 more connections than the control mean of 4,88. Also, business discussions increased by 37%. The mean index rose by 0,45 points over a control mean of 1,23.

NOTE

4.



Croatia:
Supporting Private Sector Innovation through Matchmaking between Businesses and Science

Does connecting firms with scientists and supporting their collaboration foster research partnerships and innovation?

Investment

Firms and scientists face collaboration barriers, and stronger support is needed to boost partnerships and innovation. Firms and scientists often struggle to collaborate due to information frictions, behavioral biases, uncertain transaction costs, and financial constraints. Firms rely on scientific expertise for product development or problem-solving, while scientists may need firms to commercialise their research. However, finding the right match is challenging, particularly for SMEs with limited resources for outreach. In Croatia, for instance, collaborations typically occur within university networks or through personal contacts, but emerging fields like Artificial Intelligence (AI) and Energy Efficiency might need external support.

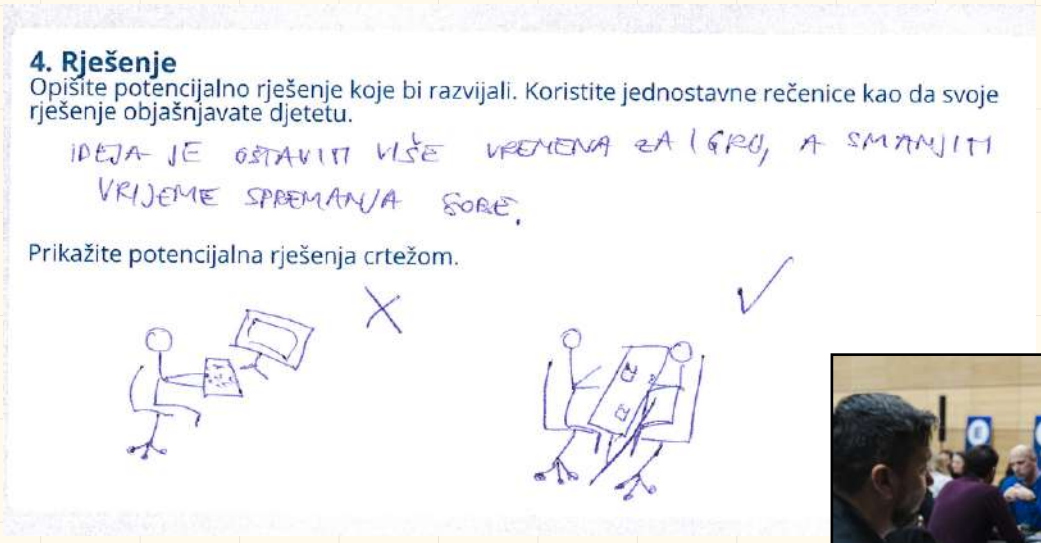
Despite recent improvements in Croatia's Research, Development, and Innovation system, challenges remain in applied research, science-business collaboration, and commercialisation of academic research. Only 8% of Croatian companies collaborate with higher education institutions, far below the EU average. To address this, the Croatian government has introduced reforms through the National Resilience and Recovery Plan and the Smart Specialisation Strategy, including increased funding for collaborative research, development, and technology transfer. However, funding alone is insufficient to foster effective science-business partnerships. For the research sector to generate innovative knowledge with economic spillover effects, and for businesses to enhance their R&D capacity, stronger collaboration between the two is vital. Matchmaking support could potentially bridge this gap by connecting firms with relevant scientists, facilitating faster identification of expertise, and coordinating meetings to overcome communication barriers and build trust.

The *Scinergy* pilot in Croatia connects businesses and scientists to foster collaborations, support innovation, and drive R&D through matchmaking events and ongoing guidance. The pilot programme was launched to strengthen business-science collaboration, particularly in high-potential fields like AI and energy efficiency. A World Bank survey revealed that 47% of companies lacked information about relevant scientists, highlighting the need for improved collaboration to drive innovation and technology transfer. To strengthen science-business collaboration, Croatia is piloting a business-science matchmaking support programme. For this purpose, the World Bank, in partnership with the Innovation Center Nikola Tesla and the Ministry of Science, Education, and Youth, designed the *"Scinergy"* pilot to facilitate science-business collaboration. *Scinergy* aims to help connect businesses and scientists based on their needs and expertise, supporting product development, joint funding applications, and public-private partnerships.

Participating firms are offered targeted connections with scientists across fields like AI, energy, and biotechnology, and are offered legal and communication support. Matched researchers and businesses first participated in a two-day matchmaking event in February 2024. The first day involved "speed-dating" sessions between firms and scientists, while the second day featured workshops for pairs interested in collaboration. These sessions encouraged ideation, with exercises like explaining potential solutions in simple terms and visualising them through drawings (Figure 4.1).[13] In March and April 2024, these pairs explored concrete R&D ideas, leading to detailed discussions and joint project planning. Pilot monitoring revealed that newly established interactions between firms and scientists continued in this period, with discussions on collaboration details, intellectual property, legal frameworks, and project financing. Firms attended specialised lectures on these topics, which deepened their understanding and aimed to facilitate cooperation.

13 A video of the event is available online: https://www.linkedin.com/feed/update/activity:7163838268210307073/

Figure 4.1: Business-Science Matching in Croatia













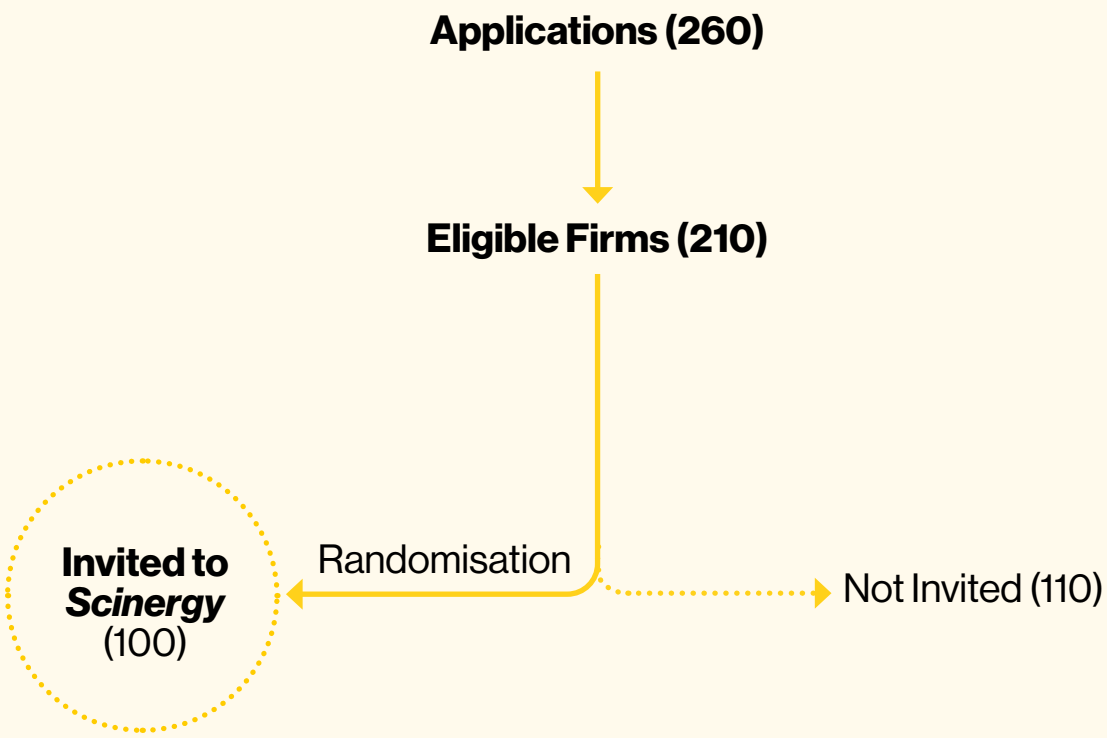
Experiment

In Croatia, the Scinergy pilot randomised firms by lottery. Applications were submitted by firms between November 16, 2023, and January 31, 2024, capturing information on scientific collaboration interests, past experiences, and future partnership preferences. Open-ended questions allowed firms to detail their R&D needs, barriers, and goals. Over 250 firms applied, with 210 qualifying for the Scinergy initiative.

Eligible businesses, regardless of size or sector, were those seeking collaboration with the scientific community.[14] Because a lottery design only holds up if the pool doesn't thin out between application and the draw, firms were asked to reconfirm their interest prior to the final selection and the launch of the event. In February 2024, 100 firms were randomly selected from the 210 eligible applicants for the business-science matchmaking pilot. These firms were matched with scientists to address their R&D needs. In other words, using a lottery, 100 out of 210 were assigned to receive matchmaking support, workshops, and post-event guidance.

14 The application data was used to match firms with scientists for the Scinergy event. If needed, the implementation team contacted the companies for further clarification.

Figure 4.2: Experimental Design in Croatia



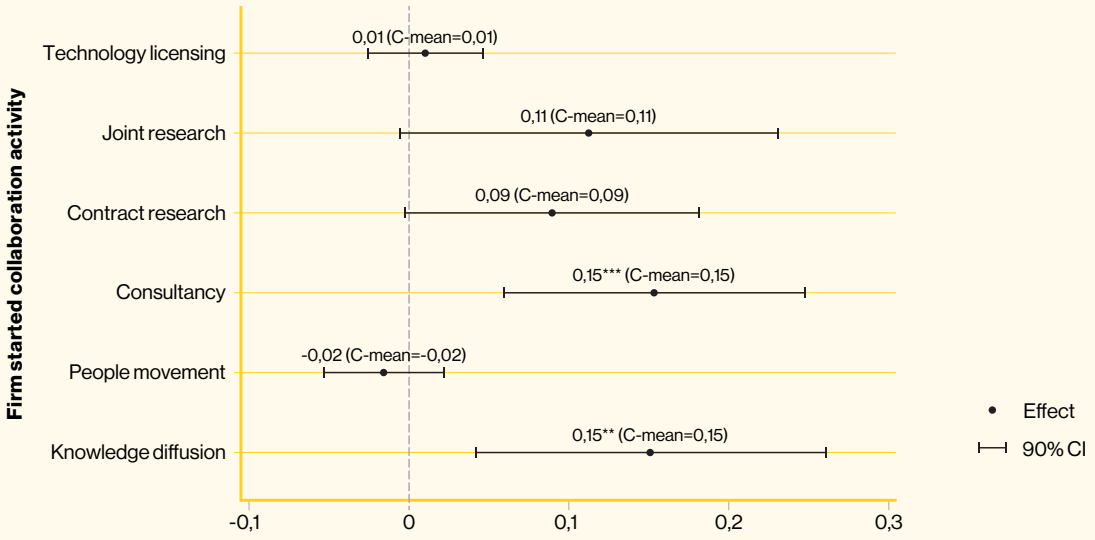
Short-Term Impacts

Collaboration attempts increased by 38% overall and ongoing or successful collaborations increased by 38%. The evaluation focused on collaboration activity, innovation outcomes, and downstream business performance. Treated firms were 18 percentage points more likely to have ongoing or successful collaborations, compared to a control mean of 47%. Across all firms, the treatment group was 22 percentage points more likely to attempt collaborations, relative to a control mean of 58%, representing a 38% increase.

Firms mostly embarked on collaborative projects with scientists to develop new technologies in the AI sector. Figure 4.3a displays what type of collaboration activities have been started. Firms in the treatment group were more likely to receive expert advice provided by academics for commercial innovation almost doubled, and to incorporate knowledge through the spreading of academic research and expertise through informal interactions, training or publications. These results show that the increase in collaborations for treated firms happened almost equally in a formal and informal manner, explaining the rather weak evidence of Scinergy on the likelihood of signing a formal collaboration agreement with a Croatian scientist. Figure 4.3b compares, between firms in the treatment and comparison group, the R&D fields in which the collaborations started since the beginning of 2024. Treated firms are more likely to have started collaborations in electrical engineering and electronics, computer science and software engineering, and AI, which is in line with the original goal of the pilot to target those sectors.

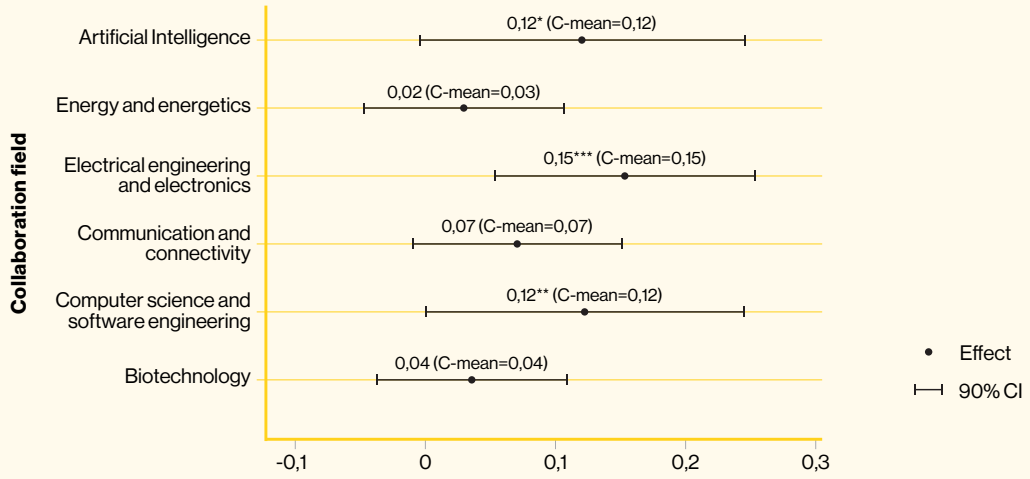
Figure 4.3: Collaboration Details

(**a**) Type of Collaboration Activity



Note: Figure 4.3a shows treatment effects (ITT estimates) on collaboration activities, derived from the survey question "Since the beginning of 2024, has your firm been starting any of the following collaboration activities with Croatian researchers?". ITT estimates are obtained from OLS regressions controlling for randomisation strata, lasso-selected firm baseline characteristics, and, when available, baseline value of the outcome variable (robust standard errors). Significance stars: *** p<0,01, ** p<0,05, and * p<0,10.
Source: Survey data (follow-up 2025 Q1) and authors' computations.

(**b**) R&D Field of Collaboration



Note: Figure 4.3b shows treatment effects (ITT estimates) on R&D fields of collaboration, derived from the survey question "In which field(s) of R&D have the collaboration activities with Croatian researchers been happening?". ITT estimates are obtained from OLS regressions controlling for randomisation strata, lasso-selected firm baseline characteristics, and, when available, baseline value of the outcome variable (robust standard errors). Significance stars: *** p<0,01, ** p<0,05, and * p<0,10.
Source: Survey data (follow-up 2025 Q1) and authors' computations.

Firms originally requesting support in AI benefited the most: they experienced a 53% increase in the completion of activities on product/process improvement, commercialisation of their new or improved products rose by 50%, and they increased research, development, and innovation (RDI) employment by over 180%. The pilot provides robust evidence that structured matchmaking support can significantly enhance collaboration and innovation, especially for firms interested in receiving support in the AI field. Treated AI-interested firms were 20,7 percentage points more likely to complete product or process improvements compared to a control group mean of 39%, representing a 53% increase. AI-interested firms in the treatment group were 25,7 percentage points more likely to commercialise new or improved products, relative to a control mean of 51%, indicating a 50% increase. Finally, treated AI-interested firms hired two more full-time RDI employees on average, compared to a control mean of 1,1, representing a 181% increase.

The economic return on investment was substantial. By reducing search costs and facilitating trust-building, the programme helped unlock the potential of Croatia's scientific and entrepreneurial ecosystems. The benefit-cost ratio is positive, especially for AI-interested firms, suggesting that every Euro invested generated more than a Euro in benefits. These findings support scaling similar interventions across other innovation-intensive sectors.

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